

Applications

IS-LM Model

Deriving IS curve

- **The following equations describe an economy:**
- $C = 10 + 0.5 Y$ (Consumption function)
- $I = 190 - 20r$ (Investment function)

Derive the equations for IS curve.

- $C=10+0.5Y$
- $I=190-20r$

$$Y = 10+0.5Y+190-20r$$

$$Y-0.5Y = 200-20r$$

$$0.5Y = 200-20r$$

$$Y = 400-40r$$

Thus, *IS* equation :

$$Y = 400-40r$$

Mathematical derivation of the LM curve

- **Given the following data about the monetary sector of the economy:**
- $M_d = 0.4 Y - 80r$
- $M_s = 1200$ mn
- where M_d is demand for money, Y is level of income, M_s is the supply of money
 - i. Derive the equation for LM function

Mathematical derivation of the LM curve Cont'd

- **Solution:**
- **For money market to be in equilibrium:**
- $M_d = M_s$
- $0.4 Y - 80 r = 1200$
- $80 r = 0.4 Y - 1200$
- $r = (0.4Y/80) - (1200/80)$
- $r = (1/200) Y - 15 \dots (i)$
- **Thus we get the following LM function:**
- $r = (1/200) Y - 15$

Applications

- **The following data is given for the monetary sector of the economy:**
- Transaction demand for money, $M_t = 0.5Y$. Speculative demand for money, $M_{sp} = 105 - 1500 i$
- Money supply $M_s = 150$
- Derive LM equation from the above data

Applications Cont'd

- **Solution:**
- Total demand function for money can be obtained by adding up the transactions demand for money (M_t) and speculative demand for money (M_{sp}). Thus
- $M_d = M_t + M_{sp}$
- $M_d = 0.5 Y + 105 - 1500r$
- In money market equilibrium

Solution

$$\begin{aligned}M_d &= M_s \\0.5Y + 105 - 1500i &= 150 \\1500i &= 0.5Y + 105 - 150 \\i &= \frac{0.5Y}{1500} - \frac{45}{1500} \\&= \frac{1}{3000}Y - \frac{3}{100}\end{aligned}$$

Thus, LM equation is :

$$i = \frac{1}{3000}Y - \frac{3}{100}$$

Alternatively, LM equation can be written as

$$\begin{aligned}Y &= 3000i + \frac{3}{100} \times 3000 \\Y &= 3000i + 90\end{aligned}$$